



BIRD

Bioinformatics, Research & Data
Bioinformatique, Recherche & Données

Bioinformatics consulting ▪ omics analysis ▪ reproducible workflows ▪ training

Bioinformatics support for your next research project

From study design to reproducible analysis, workflow development, training, and scientific mentorship.

Free 30-minute initial consultation available for new projects.

Use this meeting to discuss project goals, data type, timeline, current challenges, and next steps. This free consultation is for general orientation; detailed analysis, troubleshooting, code review, and full study design are quoted separately.

1. Study Design & Bioinformatics Consultation

Bioinformatics-oriented planning before data generation or analysis.

- Research question and analysis strategy
- Sample groups, controls, and replicates
- Metadata and sample organization
- Sequencing and downstream analysis planning
- Data management and reproducibility advice

2. Omics & NGS Data Analysis

Analysis support for genomic, omics, and high-throughput sequencing data.

- RNA-seq and differential expression
- Genomics and variant-analysis support
- PhIP-seq and peptide-enrichment analysis
- Quality control, exploratory analysis, figures, and reports
- Reproducible scripts and documentation

3. PhIP-seq & Peptide-Library Design

Specialized guidance for PhIP-seq projects and custom peptide-library planning.

- Peptide tiling strategy
- Antigen, protein, organism, or target selection
- FASTA and protein sequence preparation
- Annotation tables and control peptide planning
- Downstream enrichment-analysis strategy

4. Pipelines, Automation & Troubleshooting

Development, optimization, and troubleshooting of reproducible computational workflows.

- Python, R, Bash, and Linux troubleshooting
- Pipeline development and optimization
- FASTQ organization and data-transfer workflows
- HPC, SLURM, and Digital Research Alliance of Canada support
- README files, SOPs, and workflow handover

5. Training for Biologists & Research Teams

Customized training for students, researchers, technicians, and laboratory staff.

- Python for biologists
- R for biologists
- Basic bioinformatics and sequencing basics
- NCBI, SRA, GEO, BLAST, and public databases
- FASTA, FASTQ, BAM, VCF, and GTF/GFF files

6. Mentorship & Scientific Guidance

Technical and scientific guidance for trainees and research teams.

- Student project support
- Analysis planning and code review
- Data interpretation support
- Methods and workflow documentation

Flexible formats: free initial consultation ▪ hourly technical support ▪ fixed-price project packages ▪ customized lab workshops ▪ one-on-one training ▪ monthly lab-support plans.

Contact BIRD

Website: bird-bioinfo.com

Email: contact@bird-bioinfo.com

Location: Montréal, Québec

Insert QR code

Professional note: Services are provided independently through BIRD and can be adapted to each laboratory's needs, budget, and timeline. Institutional logos, confidential information, or unpublished project details are not used without prior authorization.

BIRD Services

Detailed service menu for laboratories and research teams

BIRD is a Montréal-based bioinformatics consulting service helping academic laboratories, research teams, students, and life sciences organizations analyze complex biological data, improve reproducibility, organize workflows, document procedures, and strengthen internal technical capacity.

Service Areas

Study design consultation	Bioinformatics-ready study planning, sample and metadata structure, controls, replicates, comparison planning, feasibility assessment, and downstream analysis strategy.
Omics and NGS analysis	Quality control, exploratory analysis, RNA-seq support, differential expression, genomics support, peptide-enrichment analysis, figures, tables, reproducible scripts, and reports.
PhIP-seq library design	Peptide tiling, antigen/target selection, FASTA and protein sequence preparation, annotation tables, control peptide planning, and enrichment-analysis strategy.
Pipeline development	Python/R/Bash workflows, automation, metadata validation, FASTQ organization, HPC/SLURM scripts, README files, SOPs, and handover documentation.
Training	Python for biologists, R for biologists, basic bioinformatics, sequencing basics, public databases, Linux/Bash, HPC, reproducibility, and research data organization.
Mentorship	Scientific and technical guidance for students, trainees, and early-career researchers, including project planning, code review, data interpretation, and workflow documentation.

Training Topics

- Python for biologists
- R for biologists
- Basic bioinformatics concepts
- Sequencing data basics
- FASTA, FASTQ, BAM, VCF, and GTF/GFF files
- NCBI, SRA, GEO, BLAST, and public databases
- Linux and Bash for researchers
- HPC, Compute Canada, and Digital Research Alliance of Canada basics
- Research data organization
- Reproducibility and documentation

Possible Service Formats

Free initial consultation	30-minute orientation meeting to discuss project goals, data type, timeline, and possible next steps.
Hourly support	Useful for troubleshooting, technical questions, code review, and short-term support.
Fixed-price packages	Useful for defined analysis projects, workflow audits, study design consultations, or training sessions.
Customized workshops	Practical training adapted to a lab, team, course, or research group.
Monthly lab support	Ongoing support for labs that need regular bioinformatics help without hiring a full-time bioinformatician.

Contact us for a free 30-minute initial consultation.

Website: bird-bioinfo.com

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Prices depend on project complexity, data volume, timeline, and required deliverables. Services can be offered as hourly support, fixed-price packages, customized workshops, or monthly support plans. Contact us for a customized estimate.